

A Multilevel Method for Time-Dependent Optimal Transport

Implementation, Replication, and Extension of Haber & Horesh (2015)

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1 Introduction and Problem Setup

Optimal transport seeks the minimal-cost way to rearrange a source measure μ_0 into a target μ_1 such that we preserve the amount of mass in any given region. Define this transport map as $T(x)$ and $u(x) = T(x) - x$

$$\min_u \int_{\Omega} |u(x)|^2 \rho_0(x) dx \quad \text{s.t.} \quad \int_{\Omega} \phi(x + u(x)) \rho_0(x) dx = \int_{\Omega} \phi(y) \rho_1(y) dy \quad \forall \phi \in C(\Omega) \quad (1)$$

For smooth densities, mass conservation forces the Jacobian constraint $\det(I_d + \nabla u) \mu_1(x + u(x)) - \mu_0(x) = 0$ (where $u(x)$ is the displacement of the point x), which is highly nonlinear and difficult to solve. Haber & Horesh instead consider the Benamou–Brenier reformulation: introduce a time variable $t \in [0, 1]$, a state variable, density $\rho(x, t) > 0$ with $\rho(\cdot, 0) = \mu_0$, $\rho(\cdot, 1) = \mu_1$, and a control variable, velocity $v(x, t)$, and minimize kinetic energy subject to the continuity equation. Substituting the momentum $m = \rho v$ convexifies the problem:

$$\min_{m, \rho} \int_0^1 \int_{\Omega} \frac{|m(x, t)|^2}{\rho(x, t)} dx dt \quad \text{s.t.} \quad \rho_t + \nabla \cdot m = 0, \quad \rho(\cdot, 0) = \mu_0, \quad \rho(\cdot, 1) = \mu_1. \quad (2)$$

Stacking space and time as $w = (m, \rho)^\top$, the constraint is a single space–time divergence $\nabla_{st} \cdot w = 0$, which has the same variational structure as the mixed form of nonlinear flow in porous media; the paper attempts to use this to re-purpose existing multigrid and saddle-point solver technology.

A point the paper stresses, and that matters for this report, is that the Benamou–Brenier formulation requires *no boundary conditions* on ρ . The continuity equation alone does not admit (or need) explicit BC on the density; the paper argues that “faking” inflow/outflow or periodic conditions, as in earlier work, biases the velocity field and degrades the reconstruction. The lack of BC for ρ instead manifests as boundary conditions on the Lagrange multipliers, resolved implicitly inside the optimization. However, as we will discuss later, relying on the optimization alone runs into some issues when it is implemented. The paper also generalizes to the L_p energy $|m|^p / \rho^{p-1}$ (strictly convex for $1 < p \leq 2$) and attempts the ill-posed L_1 limit via p -continuation. Its stated goal is a solver that is stable, handles large density contrast (where prior algorithms failed beyond a contrast of ~ 50), and has linear complexity — which it obtains by pairing a mesh-independent SQP outer loop with a multigrid linear solver.

This report is focused on the implementation and what needed to be changed/specified to actually implement it, as well as some discussion about my choices. Thus, the derivations are stated,

not developed. Section 2 discusses the paper’s decisions insofar as choosing their discretization and optimization algorithm. The substance is in Section 3 (implementation details the paper leaves unspecified, and the conditioning and filter issues that had to be resolved), Section 4 (empirical tuning), Section 5 (replication of the paper’s experiments, including the $p \rightarrow 1$ continuation, which converges here but along a qualitatively different feasibility trend than the paper reports), and above all Section 6, where three transport problems beyond the paper’s symmetric test case are analyzed.

2 Discretization and Solver Architecture

2.1 Staggered-grid discretization

A finite-volume discretization on a staggered grid places each variable where its natural flux lives. With an $n_1 \times n_2$ spatial grid and n_3 time steps:

Variable	Shape	Location
m_1	(n_1+1, n_2, n_3)	x -faces
m_2	(n_1, n_2+1, n_3)	y -faces
ρ	(n_1, n_2, n_3+1)	t -faces; $\rho[:, :, 0] = \mu_0$, $\rho[:, :, -1] = \mu_1$ fixed
λ	(n_1, n_2, n_3)	cell centers (Lagrange multiplier)

Table 1: Staggered-grid variable layout. Boundary time-faces of ρ are pinned to the source/target measures and are not optimization variables.

Boundary conditions fold into the constraint via the right-hand side q (whose nonzero entries are μ_0 and μ_1 at the first and last time slabs), so the discrete continuity equation reads $D_1 m_1 + D_2 m_2 + D_3 \rho = q$, where D_j are the directional difference operators. Crucially, the staggered placement is not cosmetic: a non-staggered (collocated) discretization of the divergence introduces null spaces and is not h -elliptic, producing the instabilities the paper attributes to earlier nodal schemes. This staggered grid is the first improvement the paper makes on previous models.

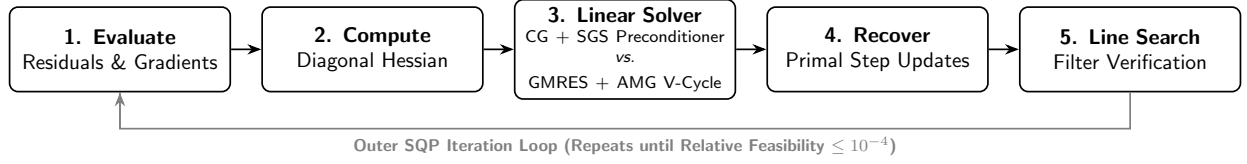
The energy integral, derived by midpoint/trapezoidal quadrature, discretizes as $h^3 (A_s m^2) A_t (1/\rho)$ with spatial/temporal averaging matrices A_s, A_t . Because the objective is pointwise but the variables live on staggered faces, the order of operations in the averaging matters tremendously: one squares the momentum then averages (otherwise an alternating $+1, -1, \dots$ field averages spuriously to zero) and inverts then averages the density. The latter guarantees $f(m, \rho) \rightarrow \infty$ as $\rho \rightarrow 0$, keeping iterates away from non-physical negative densities. This results in a more numerically stable approximation, especially for high-contrast regions.

2.2 Optimization Algorithm

The discrete problem can thus be expressed as

$$\min_{m, \rho} h^3 (A_s m^2) A_t \left(\frac{1}{\rho}\right) \quad \text{s.t.} \quad C(m, \rho) := D [m^\top; \rho^\top]^\top - q = 0. \quad (3)$$

We will elect to solve this using an inexact Newton Solver: Sequential Quadratic Programming.



The KKT conditions are solved with an inexact-Newton SQP loop. The exact Hessian of the Lagrangian has off-diagonal cross terms (from the mixed m - ρ derivative; this is also what made the original formulation difficult to solve), so the paper takes a Gauss–Newton approximation that keeps only the SPD block-diagonal $\hat{A}$, whose blocks are explicitly

$$\hat{A} = \begin{pmatrix} 2 \operatorname{diag}(A_s^\top A_t(\frac{1}{\rho})) & 0 \\ 0 & 2 \operatorname{diag}(A_t^\top A_s(m^2) \operatorname{diag}(\frac{1}{\rho^3})) \end{pmatrix}. \quad (4)$$

Each SQP step solves the symmetric-indefinite saddle-point system

$$\begin{pmatrix} \hat{A} & D^\top \\ D & 0 \end{pmatrix} \begin{pmatrix} \delta w \\ \delta \lambda \end{pmatrix} = - \begin{pmatrix} \nabla_w L \\ \nabla_\lambda L \end{pmatrix}. \quad (5)$$

Because $\hat{A}$ is diagonal it inverts trivially, and eliminating δw yields an SPD system in $\delta \lambda$ governed by the Schur complement $S = -D \hat{A}^{-1} D^\top$:

$$-S \delta \lambda = \nabla_\lambda L - D \hat{A}^{-1} \nabla_w L, \quad \delta w = -\hat{A}^{-1} (D^\top \delta \lambda + \nabla_w L). \quad (6)$$

S is a generalized space–time Laplacian $\nabla_{st} \cdot \sigma (\nabla_{st})^*$ with coefficients $\sigma = \operatorname{diag}(\hat{A}^{-1})$ that only depend on ρ and m^2 ; thus, we want to apply multigrid. Globalization uses a Fletcher–Leyffer filter, following the paper, which doubles as the inner-solver stopping rule: a step is accepted only when both the linear tolerance and the filter test are satisfied. The implementation specifics of that filter are where most of the engineering in Section 3 lives.

2.3 Numerical Solver

The paper describes two ways to solve (5); both are implemented here, and which one is used depends on the experiment (the mesh-independence test in Section 5 runs both for comparison).

gmres_amg (better for large grids, the paper’s break-even is near 64^3).

Solve the full saddle-point system matrix-free with GMRES, preconditioned by a Schur-complement block factorization: one Ruge–Stüben AMG V-cycle on S for the λ -block, then $\hat{A}^{-1}$ for the w -block (via `pyamg`). The paper notes the exact block preconditioner converges in three GMRES steps; the AMG V-cycle is what makes S^{-1} affordable.

cg_sgs (the paper’s reduced path).

Eliminate δw and solve the SPD system (6) for $\delta \lambda$ with CG preconditioned by symmetric Gauss–Seidel. The paper recommends this below the break-even mesh; here it is used for the scaling study (Section 6.4), where its non-mesh-independence is the point of interest, and as a fallback for the ring-to-disk and two-blobs cases (Section 6) where GMRES did not converge reliably.

The one substantive difference from the paper is tolerance. The paper’s AMG preconditioner delivers ~ 2 – 7 inner iterations per step at a loose inner tolerance of 0.1. SGS is not mesh-independent and needs far more iterations at that tolerance, so the `cg_sgs` path here runs a much tighter inner tolerance instead (Section 4) — trading inner work for accurate Newton directions and fewer outer steps.

3 Implementation Details and Adjustments

The paper specifies the method but not the implementation details that determine whether it converges: the exact filter bookkeeping, the floor on the Gauss–Newton blocks, the stopping rule.

3.1 Making $\hat{A}_\rho$ more stable

The paper’s ρ -block of the Gauss–Newton Hessian (4) is $2 \operatorname{diag}(A_t^\top A_s(m^2) \operatorname{diag}(\rho^{-3}))$ at $p = 2$ — it is proportional to m^2 , so it vanishes wherever the momentum is zero at a time-face. That is exactly the state at initialization, where $m = 0$. The block then collapses to its absolute floor ($\sim 10^{-12}$), and in the Schur complement $S = -D\hat{A}^{-1}D^\top$ the ρ contribution, scaling like the inverse of that block, dominates the m contribution by a factor of $\sim 10^8$. CG returns a $\delta\lambda$ inaccurate enough to *increase* the constraint violation rather than reduce it, and the filter then blocks every step from iteration 3 onward on grids $\geq 32 \times 32 \times 20$.

The fix floors $\hat{A}_\rho$ at h^3 , the scale of the m -block diagonal, which keeps the two Schur contributions within a factor of ~ 2 . The floor is only active early, while some time-faces still carry near-zero momentum; once m grows, the true ρ^{-3} values dominate and the floor is inert.

This is the same gap the two solver paths expose differently. The paper’s AMG preconditioner produces accurate CG/GMRES directions even when $\hat{A}_\rho$ is very small, so the paper never needs a floor; SGS cannot, so the `cg_sgs` path here requires it. The floor and the tightened inner tolerance (Section 4) are the two compensations for using SGS rather than AMG wherever the reduced path is used.

3.2 Filter

Three filter changes share one root cause — a Fletcher–Leyffer filter behaves degenerately when started from $m = 0$ (where $f_0 = 0$) and fed steps from a loose-tolerance inner solve.

1. **Empty initialization.** The standard seed $(f_0, h_0) = (0, h_0)$ can never be dominated, since $f_{\text{trial}} > 0$ always; acceptance then hinges solely on $h_{\text{trial}} < (1 - \gamma_h)h_0$, which fails by a hair under the small step sizes that positivity backtracking forces. Initializing the filter empty (keeping only $h_{\text{max}} = \beta h_0$) removes the degenerate entry.
2. **Relative add-threshold.** Near the optimum f oscillates by floating-point amounts; adding an entry on every $f_{\text{trial}} > f_{\text{cur}}$ builds a staircase that a 10%-tolerance CG step cannot clear. Adding only when $f_{\text{trial}} > f_{\text{cur}}(1 + 10^{-4})$ prevents the accumulation.
3. **A reverted heuristic (negative result).** An optimality-phase rule requiring $f_{\text{trial}} < f_{\text{cur}}$ once near-feasible was tried and removed: it caused 29–33 step rejections per 50-iteration run at the paper’s recommended tolerance range, preventing convergence outright.

3.3 Convergence criterion: why primal feasibility alone is not enough

The paper does not state its stopping rule explicitly, and an early version of this implementation terminated on relative primal feasibility alone, $\|C\|_1/\|C\|_1^{(0)} < \text{tol}$. That criterion is dangerously permissive here, for a reason specific to starting from $m = 0$. The objective gradient is $\partial J/\partial m = 2m/\rho$, which is *exactly zero* at $m = 0$, so the first Newton step only acts to correct the constraint. On the isotropic $42 \times 42 \times 42$ grid the Schur complement is dominated by its temporal term, which drives $\delta\lambda$ to be constant in time; since the initial constraint $C = (\mu_1 - \mu_0)/(n_3 h)$ is *also* constant in time (it is the linear interpolation residual), the exact solve of $S\delta\lambda = -C$ is constant in t and

yields $\delta\rho = 0$. After a single step the constraint is satisfied to tolerance — so a primal-only test declares victory — but ρ is still the linear interpolation $(1-t)\mu_0 + t\mu_1$, the mass-fading “crossfade,” not the Wasserstein geodesic. The visual symptom is exactly the absence of a transport trail in the intermediate density frames.

The fix restores a dual-feasibility check, with two adjustments that make it usable. First, the dual residual is measured on *interior* faces only: the boundary x/y faces have $m = 0$ pinned by the no-flux condition (Section 6.1), not by optimality, so $D^\top \lambda$ there is permanently nonzero and must be excluded. Second, because $\|\nabla_w L\| \approx 0$ at the $m = 0$ start, the normalizing scale is tracked as a running maximum over iterations, so the first genuinely nonzero value sets the scale. Convergence now requires *both* $\|C\|_1/\|C\|_1^{(0)} < \text{tol}$ and $\|\nabla_w L\|_{\text{int}}/\text{scale}_w < \text{tol}_w$, with $\text{tol}_w = 10\sqrt{\text{tol}} = 0.1$ at $\text{tol} = 10^{-4}$ — looser than the primal tolerance (the $m = 0$ start inflates scale_w), but more than tight enough to block the false one-step convergence, where $\|\nabla_w L\|_{\text{int}}/\text{scale}_w \approx 1.0$ at step 2.

4 Tuning Study

The method already converges with the literature-standard inner tolerances (0.1 for GMRES+AMG, 10^{-4} for CG+SGS), so none of the tuning below was *needed*. I ran it out of curiosity: since the inner linear solve dominates the per-step cost, I wanted to know whether the commonly used values are anywhere near optimal, and how sensitive total runtime is to them.

The result is reassuring for the literature defaults rather than a reason to deviate from them. Loosening the inner CG tolerance makes each Newton direction less accurate and costs extra outer SQP steps — at $\text{cg_tol} = 10^{-2}$ the 64^3 grid needs 42 outer iterations and ~ 100 s, versus a single outer iteration and ~ 4.6 s at 10^{-4} . Tightening past 10^{-4} , however, buys nothing: the solve is already down to one outer step there, so further tightening only adds inner CG iterations (82 at 10^{-4} rising to 108 at 5×10^{-6}) and wall-clock creeps back up. Total runtime is thus essentially monotone decreasing down to 10^{-4} and flat-to-slightly-rising below it (Figure 1, left) — the standard value sits right at the floor, so there is no faster setting to be had. The filter step cap `max_h_factor` has no measurable effect at all on these problems: across the swept range 1.05–5.0 the outer and inner counts are identical and wall-clock varies only within run-to-run noise (Figure 1, right), because the line search rarely needs to invoke the cap. The GMRES tolerance on the AMG path shows a similarly mild dependence (Figure 2): the AMG V-cycle keeps inner counts in the low teens across the whole range, so runtime barely moves.

The literature defaults are already optimal here. The sweep does not motivate deviating from the standard tolerances — it confirms them. The CG optimum sits at 10^{-4} , exactly the literature value, and the GMRES and `max_h_factor` curves are flat enough that no setting meaningfully beats the default. This is the cleanest possible outcome for impartiality: the replication (Section 5) and extension (Section 6) experiments use the standard tolerances, and the tuning study confirms that choice costs nothing rather than having to be defended against a faster tuned alternative. Had the optimum been elsewhere, using it downstream would have risked overfitting hyperparameters to one problem geometry; as it stands, the question is moot.

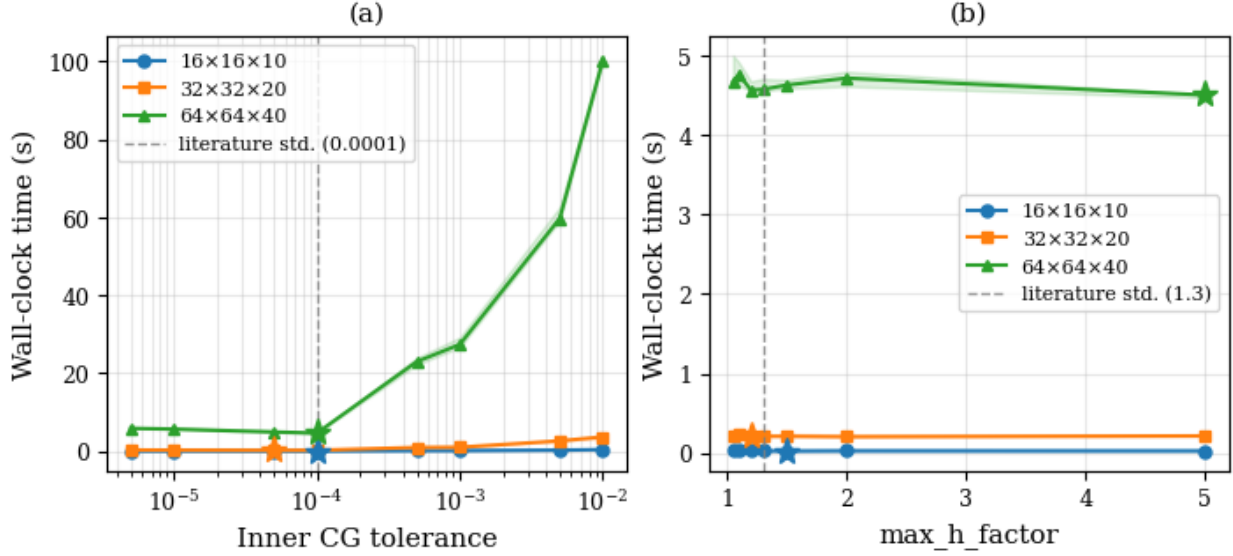


Figure 1: Wall-clock time to convergence vs. inner CG tolerance (left) and filter step cap $\max_h_factor$ (right), one curve per grid size, median of repeated runs with min-max band shaded. The dashed vertical line is the literature-standard value. Runtime falls steeply as the CG tolerance tightens to 10^{-4} and is flat to slightly rising below it; $\max_h_factor$ has essentially no effect across the swept range.

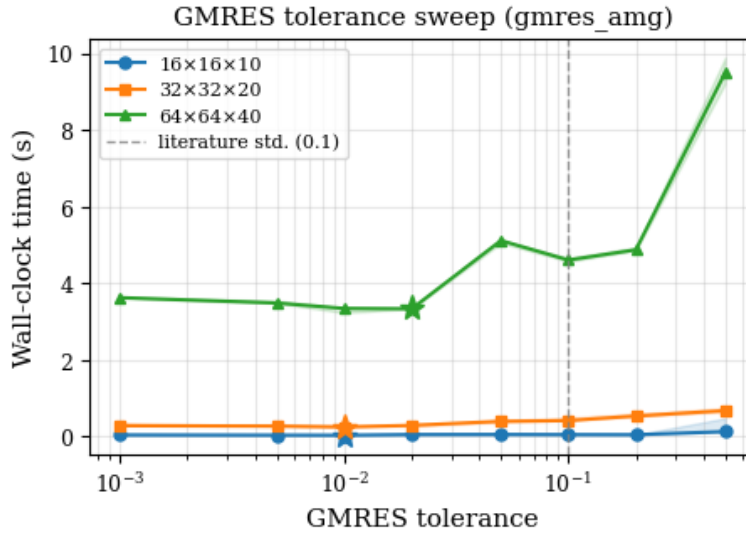


Figure 2: Wall-clock time vs. GMRES tolerance on the `gmres_amg` path, same style. The AMG preconditioner keeps the inner iteration count low across the swept range, so runtime is far less sensitive to this tolerance than to the CG one.

Parameter	Value used downstream	Notes
<code>gmres_tol</code> (inner)	0.1	literature standard; AMG path
<code>cg_tol</code> (inner)	10^{-4}	literature standard; also the tuned optimum
<code>max_h_factor</code>	1.3	filter step cap (no measurable effect)
<code>tol</code> (outer SQP)	10^{-4}	primal + dual feasibility (Section 3)
<code>filter.$\gamma_{f,h}$</code>	10^{-5}	
<code>filter.β</code>	10	$h_{\max} = \beta h_0$

Table 2: Parameters used in Sections 5 and 6. The tuning study (Figure 1) confirms the literature CG tolerance is already optimal, so no deviation is needed.

5 Replication of the Paper’s Experiments

5.1 Mesh independence (Exp. 1)

The paper’s central optimality claim is that the SQP outer-iteration count is nearly independent of mesh size — a factor-of-8 growth in unknowns between grids barely moves the count (their Table 1) — and that the GMRES+AMG inner solver shows only mild iteration growth under refinement (their Table 2). Because this implementation has both solver paths, the cleanest test is to run *both* and compare each against the paper (Table 3).

The outer SQP count is mesh-independent on either path: 6–7 iterations across the whole hierarchy on GMRES+AMG, and an essentially identical 6–7 on CG+SGS. This is the property the paper attributes to the Gauss–Newton SQP structure itself, and it holds here regardless of how the inner system is solved — the choice of preconditioner does not change how many Newton steps the outer loop needs. The counts are lower than the paper’s 16/13/14 because the inner solve here is run to convergence rather than filter-gated, so each step is more accurate, but the flatness across grids is the same.

In contrast, the inner behavior is where the two paths separate, and it is exactly the separation the paper’s “break-even” discussion predicts. On GMRES+AMG the inner count stays nearly mesh-independent: the first iteration of every solve needs just 2 inner steps regardless of grid, and the mid/late counts rise only gently from 7–7 at 16^3 to 11–15 at 64^3 — mirroring the paper’s own 2/3/2 through 2/6/7. On CG+SGS the inner count grows much faster, $24 \rightarrow 44 \rightarrow 82$ on the first iteration, because symmetric Gauss–Seidel is not a mesh-independent preconditioner. Both paths converge to the same solution in the same handful of outer steps; AMG simply keeps the per-step linear-solve cost flat under refinement where SGS does not. This is precisely why the paper reaches for AMG on large grids, and it is the same SGS limit the scaling study (Section 6.4) later quantifies.

Grid	Paper (GMRES+AMG)		This work, GMRES+AMG		This work, CG+SGS	
	SQP	inner (1/mid/last)	SQP	inner (1/mid/last)	SQP	inner (1/mid/last)
$16 \times 16 \times 10$	16	2 / 3 / 2	6	2 / 7 / 7	6	24 / 17 / 16
$32 \times 32 \times 20$	13	2 / 5 / 3	6	2 / 9 / 10	6	44 / 31 / 31
$64 \times 64 \times 40$	14	2 / 6 / 7	7	2 / 11 / 15	7	82 / 60 / 60

Table 3: Mesh independence, $p = 2$, density contrast 10. “inner” is GMRES iterations for the AMG columns and CG iterations for the SGS column, reported at the first, a middle, and the last SQP iteration. The outer SQP count is mesh-independent on both paths (6–7); the inner count stays nearly flat with AMG but grows with the mesh under SGS, matching the paper’s break-even rationale.

5.2 Density evolution at contrast 10 (Exp. 2)

The paper’s first qualitative result (its Fig. 3) solves the four-quarter-circle problem at $p = 2$ and contrast 10 on the isotropic $42 \times 42 \times 42$ grid and displays the resulting density evolution. Figure 3 reproduces it: the four corner blobs route inward along the diagonals into the centered disk, with the intermediate frames showing a genuine transport trail rather than mass fading in place. This is precisely the grid on which a primal-only stopping rule silently accepts the linear-interpolation crossfade (Section 3), so the presence of the trail here doubles as a visual confirmation that the dual-feasibility check is doing its job.

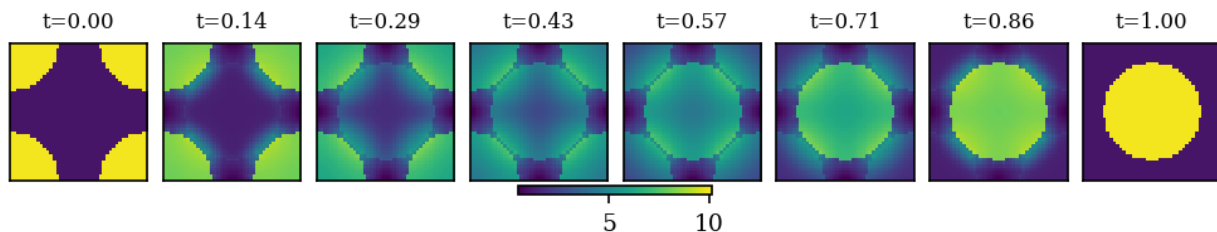


Figure 3: Density evolution $\rho(\cdot, t)$ at eight evenly spaced time slices for Exp. 2: $p = 2$, contrast 10, $42 \times 42 \times 42$ grid (the setup of the paper’s Fig. 3). The four corner blobs of μ_0 transport into the single centered disk of μ_1 , routing inward along the diagonals rather than fading in place.

5.3 High density contrast, multilevel (Exp. 3)

The paper stresses that no prior algorithm known to the authors solved transport at density contrast above ~ 50 ; their multilevel scheme handles contrast 100. Their Table 3 reports SQP counts of 28 / 12 / 14 on the $16^2 \times 8 \rightarrow 32^2 \times 16 \rightarrow 64^2 \times 32$ hierarchy. This implementation solves the same contrast-100 problem with comparable behavior (Table 4): the coarse level takes the most work (13 SQP iterations) and each finer level, warm-started from the interpolated coarse solution, converges in just 4. The pattern matches the paper’s — most of the effort is spent on the coarse grid and the multilevel warm start makes the fine grids cheap — and the high-contrast case is handled stably, with no blow-up. Figure 4 shows the density evolution at the finest level and Figure 5 the constraint history across the hierarchy.

Part of the coarse-grid count deserves a caveat. The time spent on the $16^2 \times 8$ level is inflated by the stopping rule rather than by genuine optimization difficulty: the convergence test of Section 3

couples primal and dual feasibility, and the pair is hard to tune jointly — in the other experiments at most one of the two measures was ever the suboptimal one, and the setting used throughout is the best compromise found. On this coarse grid the combined test keeps the loop alive after the constraint violation has largely settled, waiting for the dual measure to clear its threshold, which accounts for a meaningful share of the 13 iterations.

Grid	Paper SQP iters	This work SQP iters
$16 \times 16 \times 8$	28	13
$32 \times 32 \times 16$	12	4
$64 \times 64 \times 32$	14	4

Table 4: High-contrast multilevel, $p = 2$, contrast 100 (paper Table 3). Both show the coarse level dominating the work and the warm-started fine levels converging quickly; the coarse-grid count here is partly an artifact of the coupled primal–dual stopping rule.

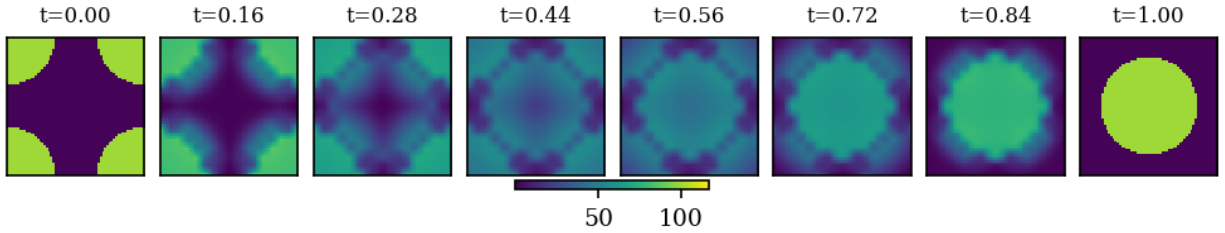


Figure 4: Density evolution $\rho(\cdot, t)$ at eight evenly spaced time slices for Exp. 3: contrast 100 at the finest multilevel level. The transport remains stable and routes mass through the interior despite the hundredfold density contrast.

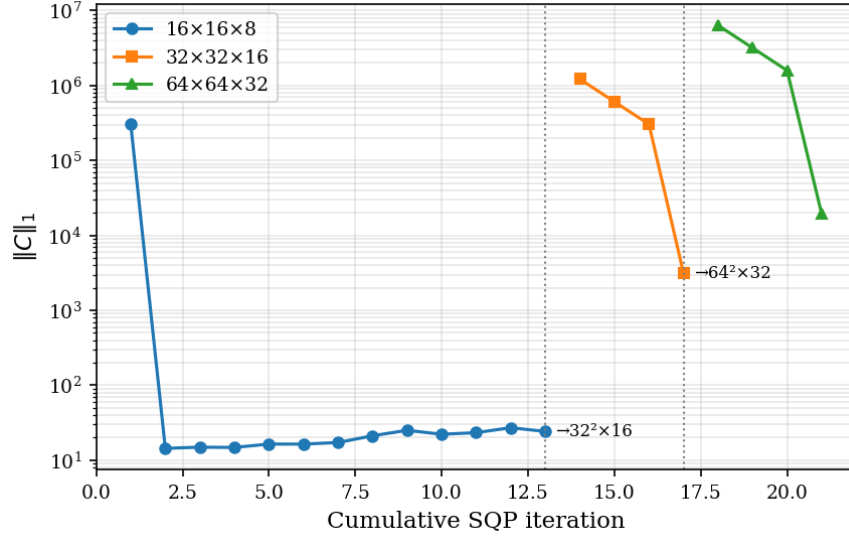


Figure 5: Constraint violation $\|C\|_1$ versus cumulative SQP iteration across the multilevel hierarchy (Exp. 3), each level in a distinct color with a dotted line at each level transition. The warm start from the coarser level keeps the feasibility drop steep on each successive grid.

5.4 The $p \rightarrow 1$ continuation (Exp. 5)

The paper’s final experiment relaxes p from 2 toward 1 by simple continuation (100 steps to $p = 1.01$, one SQP iteration per p , on a $64^2 \times 32$ grid at contrast 10) and reports that both primal and dual feasibility decrease smoothly as $p \rightarrow 1$, concluding that this “was sufficient in order to obtain an approximation to the L_1 problem.” This implementation reproduces the headline result: the continuation converges, with feasibility holding near 10^{-5} throughout — well below the 10^{-4} tolerance — so a usable approximation to the L_1 problem is obtained. What does not reproduce is the trend. Rather than decreasing monotonically, feasibility here traces a shallow U (Figure 6), falling through the middle of the continuation and turning upward again as p approaches 1. The mechanism behind the upturn is understood: the $\rho^{-3} \rightarrow \rho^{-(p+1)}$ growth in the ρ -block (4) as p falls worsens the conditioning problem of Section 3.1, and the h^3 floor on that block, chosen for $p = 2$, is not re-scaled as p decreases. On this problem neither effect ever threatens the tolerance, so there is no cause for alarm — but the trend leaves clear room for improvement. A p -dependent floor and a preconditioner that tracks the worsening conditioning are the natural next steps, and on harder problems (finer grids, higher contrast) the upturn could plausibly grow into a genuine obstruction, making the low- p regime the least robust corner of an otherwise well-behaved solver.

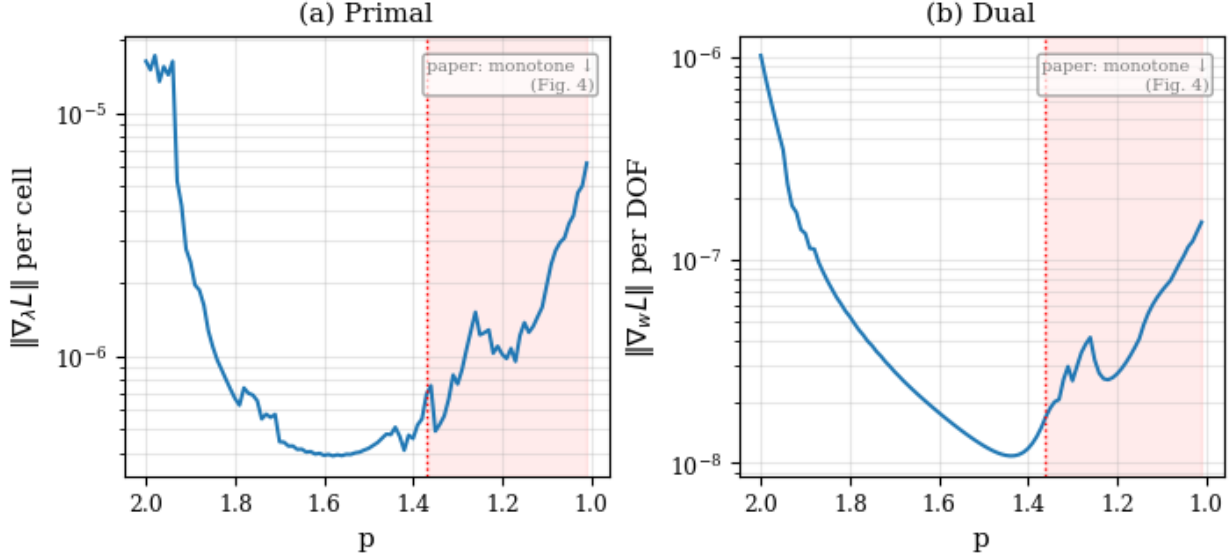


Figure 6: Primal (left) and dual (right) feasibility as p decreases from 2 to 1.01 (inverted x -axis, so the approach to $p = 1$ is rightward; Exp. 5). Feasibility traces a shallow U rather than the paper’s monotone decrease, but remains near 10^{-5} — well below the 10^{-4} tolerance — throughout, so the continuation converges and yields a usable L_1 approximation. The shaded region marks the late upturn, the conditioning headroom discussed in the text.

6 Extended Experiments — New Transport Problems

Beyond the paper’s symmetric test case, three transport problems were constructed to probe the solver under asymmetry, topology change, and mass merging. Each is run on the $16 \times 16 \times 10$ and $32 \times 32 \times 20$ grids to test whether mesh independence survives harder geometry (the scaling study of Section 6.4 extends the sweep to $48 \times 48 \times 30$), with density time-slice strips taken from the $32 \times 32 \times 20$ run. The asymmetric case is run on the `gmres_amg` path; the ring-to-disk and two-blobs cases use `cg_sgs`, since GMRES did not converge reliably across all grid sizes on those two geometries — itself a small finding about where the AMG path’s robustness gives out.

6.1 Asymmetric Gaussians (translation + rescaling)

A wide Gaussian centered at $(0.25, 0.25)$ transports to a narrow Gaussian at $(0.75, 0.75)$ — a diagonal translation combined with a variance change, with no symmetry to cancel boundary effects.

With the no-flux fix in place (below), the asymmetric case converges on both grids. Figure 7 shows the resulting density evolution (top) and the constraint history (bottom). One result worth flagging: unlike the symmetric replication, the outer-iteration count here grows with refinement — roughly 58 iterations on $16 \times 16 \times 10$ against roughly 100 on $32 \times 32 \times 20$, with the finer run’s primal residual passing the relative threshold long before termination. Mesh independence, in other words, is partly a property of the problem geometry and not of the method alone, and the coupled primal–dual stopping rule (Section 3) is again what stretches the tail.

Boundary flux: diagnosed and fixed. This case first exposed a property of the paper’s formulation that its own test problem hides. The paper imposes *no* boundary conditions (Section 1);

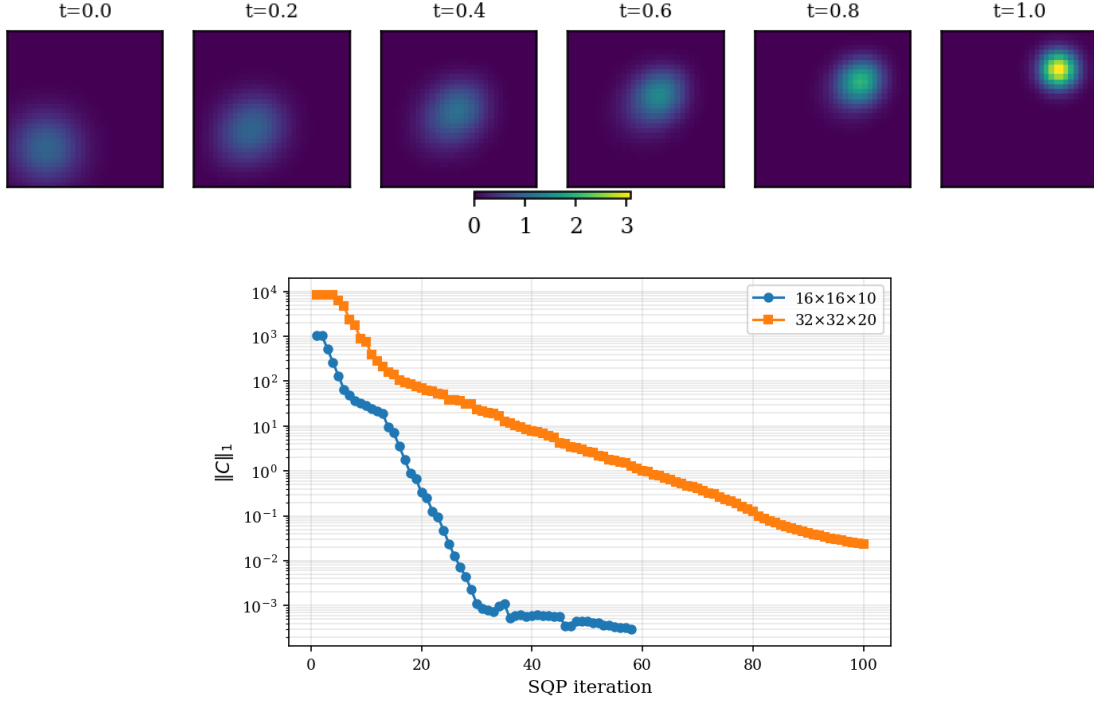


Figure 7: Top: Asymmetric Gaussian density evolution ($32 \times 32 \times 20$, no-flux boundaries) — wide Gaussian at $(0.25, 0.25)$ transported to a narrow one at $(0.75, 0.75)$. Bottom: $\|C\|_1$ versus SQP iteration under the no-flux fix. The curves do not cluster (roughly 58 vs. 100 outer iterations), so mesh independence does not fully carry over to this asymmetric geometry.

zero boundary flux is only energetically favorable, not enforced, and the staggered divergence treats the boundary momentum faces $m_1[0], m_1[-1], m_2[:, 0], m_2[:, -1]$ as free variables. On the paper’s symmetric four-quarter-circle case this is harmless — equal mass enters and exits each wall, so the leakage cancels — but the asymmetric bottom-left $\rightarrow$ top-right transport has a genuine diagonal pull with nothing to cancel it, and the optimizer routed mass through the domain walls when that lowered the Lagrangian. Mean absolute boundary flux measured ~ 0.07 (momentum units) on the 16^2 grid before the fix.

The fix pins those boundary momentum faces to zero, treating them as constants rather than optimization variables, which enforces a hard no-flux condition $m \cdot n = 0$ on $\partial\Omega$. This is a deliberate addition *beyond* the paper, which omits such conditions by design; it removes the leakage on asymmetric problems while leaving the symmetric replication results unchanged (where the flux was already negligible). With the faces pinned, the asymmetric case transports cleanly through the interior.

The contrast is clearest in the convergence behavior. Under the original free boundaries the coarse grid still reaches feasibility — but only by routing mass through the walls (the ~ 0.07 boundary flux above) — while the $32 \times 32 \times 20$ run stalls near $\|C\|_1 \approx 10^{-1}$ and never meets the combined stopping criterion (Figure 8b). With the faces pinned, both grids descend to convergence (Figure 7). Figure 8a overlays the two directly and shows a telling detail: the free-boundary run actually descends *faster* at first, because leaking mass through the walls is the cheapest way to reduce the constraint violation — before flatlining once that shortcut is exhausted. The before/after density strips (Figure 8 vs. Figure 7) show the mass that the free-boundary run bled toward the

walls being retained in the interior once the condition is enforced.

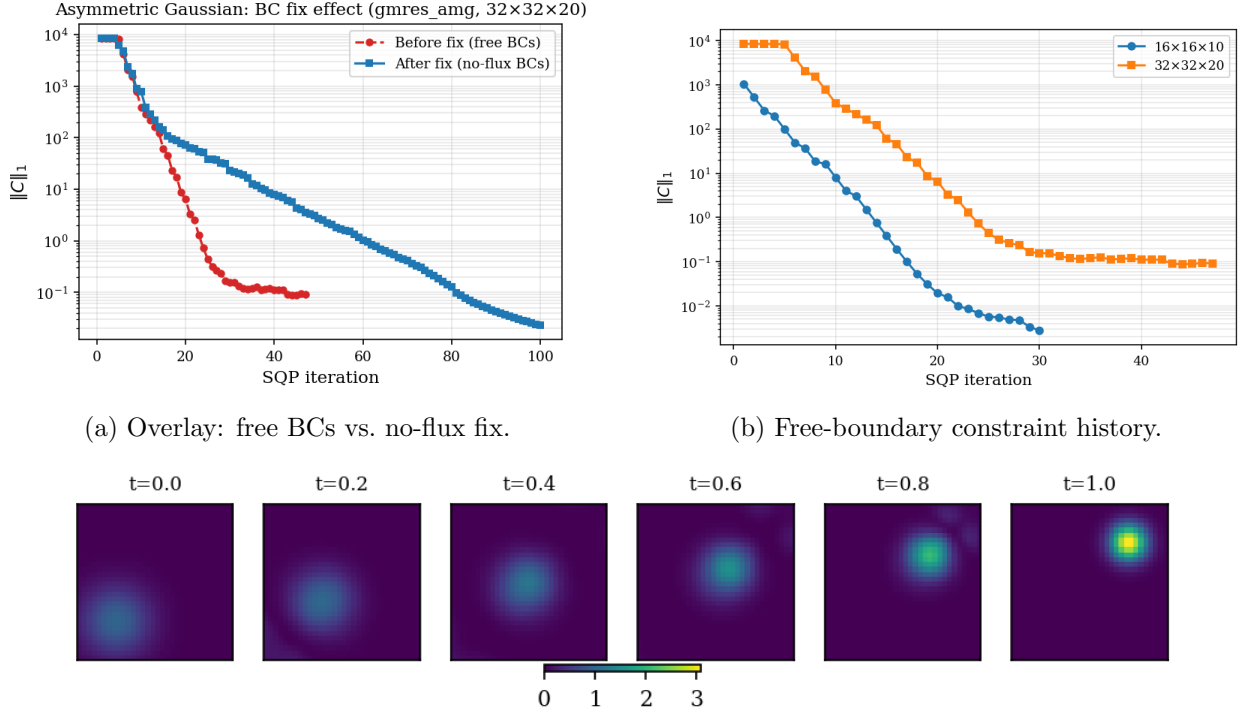


Figure 8: Top left: free-boundary vs. no-flux convergence on the $32 \times 32 \times 20$ grid — the free-boundary run descends faster initially (wall leakage is the cheap route) but stalls near 10^{-1} . Top right: per-grid history before the fix; the $32 \times 32 \times 20$ run flatlines near $\|C\|_1 \approx 10^{-1}$ and never converges. Bottom: pre-fix density strip showing faint boundary artifacts as mass drifts toward the walls.

6.2 Ring to disk (topology change)

An annular ring collapses to a central disk, forcing mass inward radially — a change in the support's topology that stresses the continuity constraint.

The topology change turns out to be handled stably: on both grids the constraint violation descends near-geometrically with no stalls or visible step rejections (Figure 9). The cost shows up instead in the outer count, which roughly doubles with refinement — about 16 iterations on $16 \times 16 \times 10$ against about 37 on $32 \times 32 \times 20$ — a milder version of the mesh dependence seen on the asymmetric case, but again short of the replication-grade mesh independence of the symmetric problem.

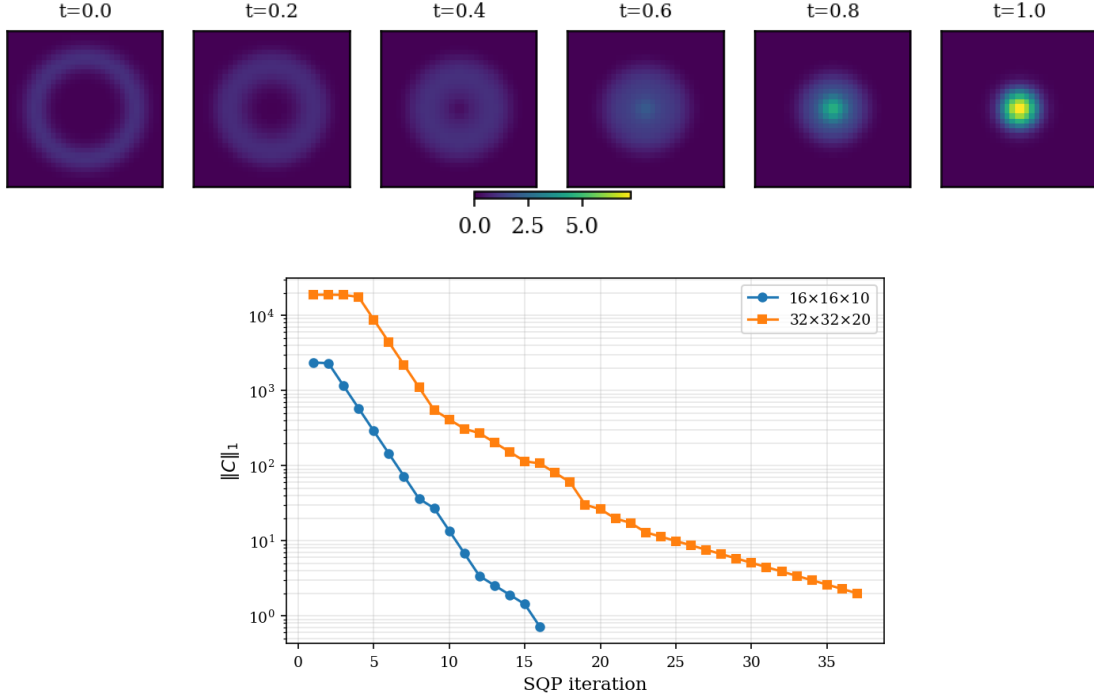


Figure 9: Ring-to-disk (`cg_sgs`, no-flux boundaries): density strip (top) as the annular ring collapses radially inward, and $\|C\|_1$ versus SQP iteration (bottom). Both grids descend steadily; the outer count roughly doubles with refinement (16 to 37 iterations).

6.3 Two blobs to one (mass merging)

Two separated blobs merge into a single central Gaussian. Consistent with its geometry — interior-only flow with no topology change — this is the best-behaved of the three new cases, and on the coarse grid it is the fastest solve in this report: the $16 \times 16 \times 10$ run converges in just 4 outer iterations, dropping the constraint violation by nearly five orders of magnitude in its first three steps (Figure 10). The $32 \times 32 \times 20$ run spends its first ~ 10 iterations on a plateau — consistent with the small early steps that positivity backtracking and the filter force from the $m = 0$ start — before entering a clean geometric descent and converging by iteration 25. Neither the asymmetric case’s boundary pathology nor the ring’s doubled outer count appears here.

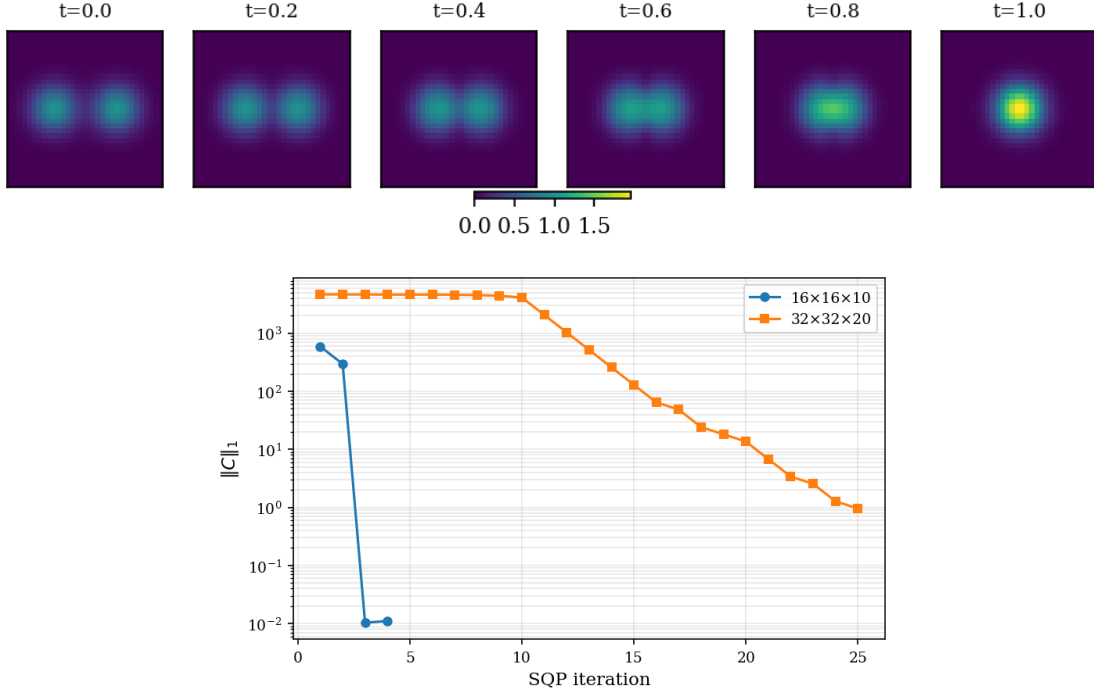


Figure 10: Two-blobs-to-center (`cg_sgs`): density strip (top) as two separated Gaussians merge into one, and $\|C\|_1$ versus SQP iteration (bottom). The coarse grid converges in 4 iterations; the finer grid plateaus for ~ 10 iterations before a clean descent to convergence by iteration 25.

6.4 Scaling / complexity study

A central claim of the multilevel approach is near-linear complexity in the number of unknowns. Figure 11 plots, on log-log axes, both the total inner-iteration count and the wall-clock time of the `cg_sgs` solver against problem size, swept over four grids from $16 \times 16 \times 10$ ($\sim 9.7\text{k}$ unknowns, 2.4s) up to $48 \times 48 \times 30$ ($\sim 271\text{k}$ unknowns, 743s). The inner-iteration count is the cleaner measure: it scales almost exactly linearly, with a four-point fit of $\alpha \approx 0.95$ (Figure 11a), reflecting the flat SQP outer count combined with only mild per-step inner growth (1789, 4285, 5569 inner iterations on the first three grids).

Wall-clock time tells a rougher story because of the finest grid. Over the first three grids it scales near-linearly — a fitted exponent of $\alpha \approx 1.22$, close to the inner-iteration trend — but the 48^3 run hit the inner-iteration budget cap (50,000 total CG iterations, the top-right point of Figure 11a), and its disproportionate runtime pushes the four-point wall-clock fit up to $\alpha \approx 1.68$ (Figure 11b). This is the SGS-versus-AMG limit of Section 2.3 made quantitative: because the SGS-preconditioned inner solver is not mesh-independent, inner iterations eventually grow fast enough to dominate, and on the largest grid the solver runs out of its iteration budget before fully converging. The near-linear regime is real but holds only while the inner solver keeps up; an AMG preconditioner on this path would be needed to sustain linear complexity to larger grids, exactly as the paper’s design anticipates.

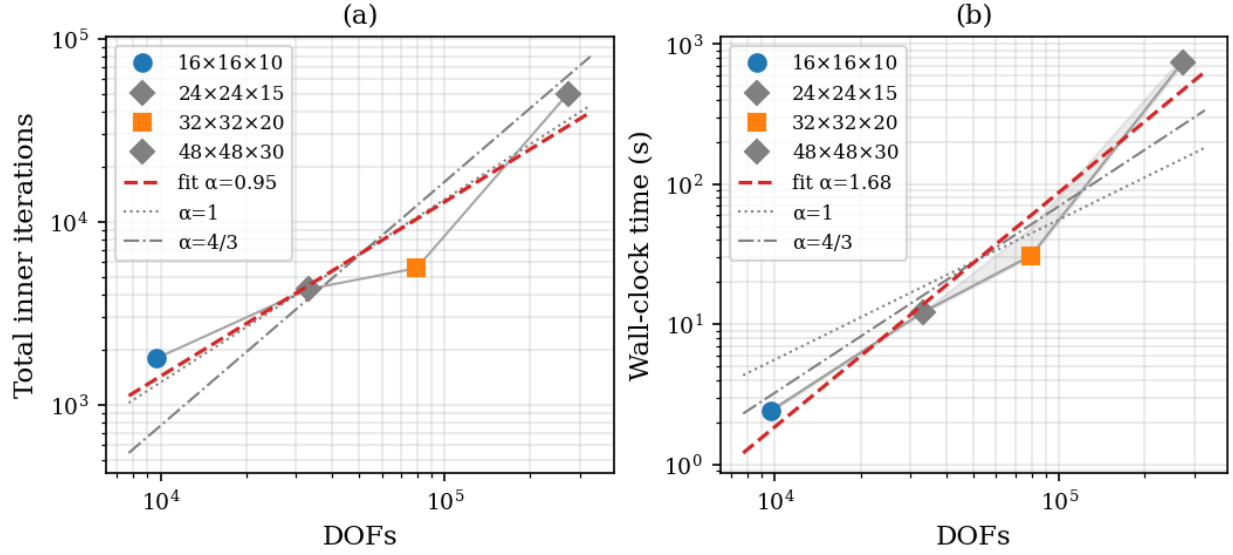


Figure 11: Scaling of the `cg_sgs` solver on log-log axes, with $\alpha = 1$ and $\alpha = 4/3$ reference slopes. (a) Total inner CG iterations versus problem size: near-linear ($\alpha \approx 0.95$ four-point fit), with the 48^3 run at the 50,000-iteration budget cap. (b) Wall-clock time versus problem size: near-linear over the first three grids ($\alpha \approx 1.22$), but the capped 48^3 point inflates the four-point fit to $\alpha \approx 1.68$, reflecting the non-mesh-independence of the SGS preconditioner.

7 Conclusion

This work reimplemented the Haber–Horesh scheme from scratch and tested it on both the paper’s own experiments and three transport problems outside its symmetric test case. The paper’s central claims replicate. The SQP outer loop is mesh-independent on the symmetric problem — 6–7 iterations across an $8\times$ growth in unknowns — and the contrast-100 case, beyond the failure point the paper attributes to all prior algorithms, solves stably, with the multilevel warm start concentrating the work on the coarsest grid (13 SQP iterations there against 4 on each finer level, and part of even that count an artifact of the coupled primal–dual stopping rule rather than of the optimization). The $p \rightarrow 1$ continuation also converges, holding feasibility near 10^{-5} , comfortably below tolerance, though along a shallow U rather than the paper’s monotone decrease; the late upturn traces to the conditioning of the ρ -block at low p and marks the least robust corner of the solver.

The extended experiments sharpen that picture in two ways. First, mesh independence turns out to be partly a property of the problem geometry, not of the method alone: outer counts grow with refinement on all three new problems — mildly for the interior-only two-blobs merge (4 to 25 iterations, the coarse-grid run being the fastest solve in this report), by roughly a factor of two for the ring-to-disk topology change (16 to 37), and most for the asymmetric Gaussian (about 58 to about 100). Second, the asymmetric case exposes the structural blind spot the paper’s symmetric test never forces: with no boundary conditions imposed, the optimizer routes mass through the domain walls, and the free-boundary run even descends faster at first — leakage is the cheap route to feasibility — before stalling near 10^{-1} on the finer grid. Pinning the boundary momentum faces to zero, a deliberate addition beyond the paper, is the difference between that stalled solve and a convergent one. Topology change and mass merging, by contrast, required no intervention at all.

The cost side of the ledger matches the paper’s design analysis. With SGS in place of AMG, the inner solver is not mesh-independent (CG counts of $24 \rightarrow 44 \rightarrow 82$ on the replication hierarchy), and the scaling study quantifies where that bites: near-linear wall-clock growth ($\alpha \approx 1.22$) over the first three grids gives way to budget exhaustion at 48^3 , inflating the four-point fit to $\alpha \approx 1.68$. The tuning study, finally, confirms rather than revises the literature defaults — the inner CG optimum sits exactly at the standard 10^{-4} , and the remaining hyperparameters are flat across their swept ranges. The natural next steps follow directly from the failure modes observed: an AMG preconditioner on the reduced path to sustain linear complexity, a p -dependent floor for the low- p regime, and a more principled treatment of the coupled primal–dual stopping rule, which is the common thread behind the inflated coarse-grid count at contrast 100 and the long convergence tails on the asymmetric problem.

References

- [1] E. Haber and R. Horesh, *A Multilevel Method for the Solution of Time Dependent Optimal Transport*, Numer. Math. Theor. Meth. Appl. **8**(1), 97–111, 2015. <https://doi.org/10.4208/nmtma.2015.w02si>